

Weekly Progress Report – Week 2

Intern Name: Anshika Kodukula

Organization: Trinetra Drones and Robotics Pvt. Ltd.

Week: Week 2

Reporting Period: June 8, 2026 – June 14, 2026

Objectives

- Identify agricultural datasets for crop disease detection.
- Study image processing and machine learning concepts.
- Learn Python basics for AI development.
- Understand applications of AI in agriculture.
- Initiate digital marketing activities for the organization.

Activities Performed

Dataset Collection

- Researched publicly available agricultural datasets.
- Studied PlantVillage, Rice Leaf Disease, Tomato Leaf Disease, and Corn Leaf Disease datasets.
- Documented dataset features and applications.

Study of AI Technologies

- Learned image processing concepts including image acquisition, enhancement, segmentation, and feature extraction.
- Studied machine learning concepts such as training, testing, and classification.
- Explored the fundamentals of Convolutional Neural Networks (CNNs).

Python Learning

- Practiced Python basics including variables, loops, functions, and importing libraries.
- Explored introductory concepts related to NumPy and OpenCV.

AI in Agriculture Research

- Studied AI applications in crop disease detection, crop monitoring, precision agriculture, and yield prediction.
- Analyzed the role of drones in smart farming.

Skills Acquired

- Understanding of agricultural datasets.
- Basic image processing concepts.
- Introduction to machine learning and deep learning.
- Basic Python programming skills.
- Knowledge of AI applications in agriculture.

Challenges Faced

- Understanding the structure of agricultural datasets.
- Learning new AI and machine learning concepts.

Outcome

Successfully completed research on agricultural datasets and AI technologies used in modern farming. Developed foundational knowledge required for future AI-based agricultural applications and drone-assisted farming solutions.

Conclusion

Week 2 activities helped build a strong foundation in AI, image processing, machine learning, and agricultural datasets. The acquired knowledge will support future project activities involving crop monitoring, disease detection, and smart agriculture solutions.